Higher National Diploma in Engineering (QS)
2nd Year 1st semester Examination- 2019
QS 21054 Civil Engineering Construction 11 for QS

Instructions for Candidates:
Answer any 5 questions.

No. of questions : 06
No. of pages : 07
Time : 3 hours

Q1)"Port City" construction is one of the biggest critical ongoing project in Sri Lanka. Once the project city is completed, it adds 269 hectares to Sri Lankan map. In that they have proposed to build 67 story building and several other buildings. Whole area is built by filling the sea using sea sand.

(a) What type of a rise building, which has 67 stories? (02 marks)

(b) What are the special characteristics which should be attended in planning the building? (02 marks)

(c) State the reason to pay a special attention on foundation of that building? (03 marks)

(d) What type of cement do you propose for the construction of coastal belt and justify you answer. (03 marks)

(e) As a junior QS assigned to be under that project, how would you ensure that cement, satisfies the required quality and standard? (04 marks)

(f) When preparing the BOQ of that building it was need to calculate the mass of cement requires for plastering of 18 typical stories in one story whose internal dimension is 8m×12m. And floor to floor height is 3.3m. Thickness of plastering is 12mm and assumes only walls and soffit should be plastered, and ratio of cement to sand is 1: 4.

To convert wet volume of mortar to dry volume of mortar multiply by 1.35.

To be plastered in each floor 10% of whole area to be plastered is removed for ventilation.

Add a 5% of wastage.

Take Density of cement as 1440 kg/m^3 .
(06 marks)

Q2) In the above construction in the Port City, they are planning to build the 57 story building with typical floors.

a. What type of form work should be adopted for these typical floors? (03 marks)

b. One technical officer suggests to have timber from work. Do you agree with his conception? If Yes/No, justify your answer. (05 marks)

c. What are factors that a contractor should be considered when erecting form work at an environment like port city? (06 marks)

d. Discuss the reasons to have from work failures. (06 marks)

Q3) Earlier it was planned to use pre-stressed concrete locks for the building. But it was later rejected due to unavailability of a good pre-stressed concrete yard. As a substitute it was decided to use ready mix concrete.

a. Write two advantages of using pre-stressed concrete than of ready mix concrete? (2 marks)

b. How to pump concrete when upper floors are erecting? (4 marks)

c. If you are the person in charge for the concrete, what are information you would provide for the supplier when ordering the pre-mixed concrete?(6 marks)

d. Discuss four factors which the strength and durability is affected by? (8 marks)

Q4) It has been found from an initial study that the 28 days compressive strength of concrete at W/C ratio of 0.5 for a given crushed coarse aggregate (20mm maximum aggregate size) and the sea sand (naturally weathered) is49 MPa. Percentage passing the 600 sieve as a percentage of total fine aggregate content is found to be 60% for the sea sand. Consider the specific gravity of the coarse and fine aggregate is 2.65.

a) Calculate target strength for Grade25 concrete.

Note: - Target strength is the mean strength of the concrete at which no more than 5% of test specimen fall below specified characteristic strength of concrete.

Considering the general variability of the concrete mixing and the materials, if is safe to assume standard deviation of the concrete mix to be 4 MPa.

Compressive strength of concrete cube test is assumed to 3 standard normal distribution and the value of 95% confidence interval for standard normal distribution is equal to 1.64. (4 marks)

b) Find mix proportions for the calculated target strength in part (a) for a required Slump of between 60-180mm. For this calculation assume that both the fine and coarse aggregate are in saturated surface dry condition (SSD)

Note: - Following table, charts and instructions extracted from the British method of mix section are provided.

Trial water contents for the different workability requirements are given, Trend of Charge of compressive strength with water/cement ratio is given, Variation of fresh of concrete density against water content for different values of specific density of fine and coarse aggregate mix is shown and content of fine aggregate to be used in the mix as a percentage of total aggregate. This depends on the water/cement ratio, workability requirement, maximum size of aggregate and fineness ratio of fine aggregate used in the mix.

Water content of the mixed aggregate (crushed and uncrushed) should be calculated as1/3 of the water requirement of the coarse aggregate and 2/3 of the water requirement of the fine aggregate.

(5 marks)

c) For a given day it is found that the natural moisture content is 1% for coarse aggregate and 2% for fine aggregate (sea sand). Assuming that the moisture absorption for the SSD condition of the two aggregate, coarse and fine, is 0.5% and 6% respectively, find the adjusted mix proportions. (03 marks)

d) Assume that the above mix selection has addition constraints to ensure that the cement content of the mix to be between 400kg/m^3 and 500kg/m^3 and that the minimum W/C ratio of the mix to be 0.5.

I. Based on the calculation above part Q2. (b) Describe what limitations you would face now in selecting concrete mix for the above designation under the above constrains. (4 marks)

II. Suggest mix proportions based on the above limitations and discuss what implication your selection will have on the desirable strength and workability limits of the original mix. (4 marks)

Q5)

a) Do you believe building service engineering is important for a building? If so state the reason. (2 marks)

b) What are the aspects that falls under building service engineering? (2 marks)

c) It was needed to design a water distribution system for the above building for gardening. What is the appropriate diameter of the pipe it the vertical head is 4m, flow rate is 1 liter/s and effective pipe length is20m. (4 marks)

d) What are the important protection factors to be considered when designing fire safety system for the above building? (4 marks)

e) Discuses fire protection devices and explain the importance of these and select the most suitable fire device to this building. (8 marks)

Q6) Finishing of the above building is very important.

a) What are the masonry units that can be adopted for the construction? (3 marks)

b) What are various type of bonds used generally in brick masonry? (2 marks)

c) What are the objectives & requirement of plastering? (4 marks)

d) This part city is subjected to a heavy rain in same durations of the year. What type of roof you will suggest for this area. Why? (3 marks)

e) In order to be a sustainable building. What are structures/modifications you will proposed and discuss their suitability? (8 marks)

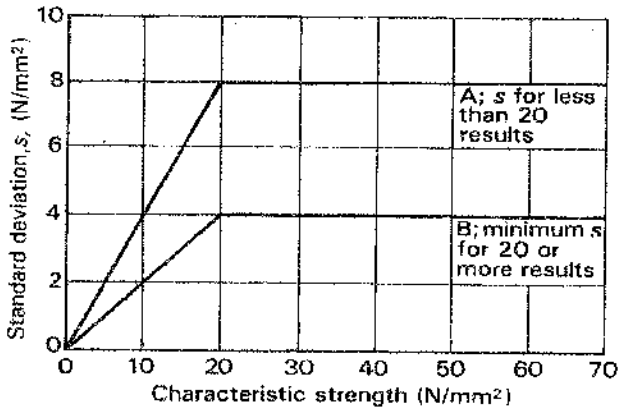

Figure 1
Relationship between standard deviation and characteristic strength

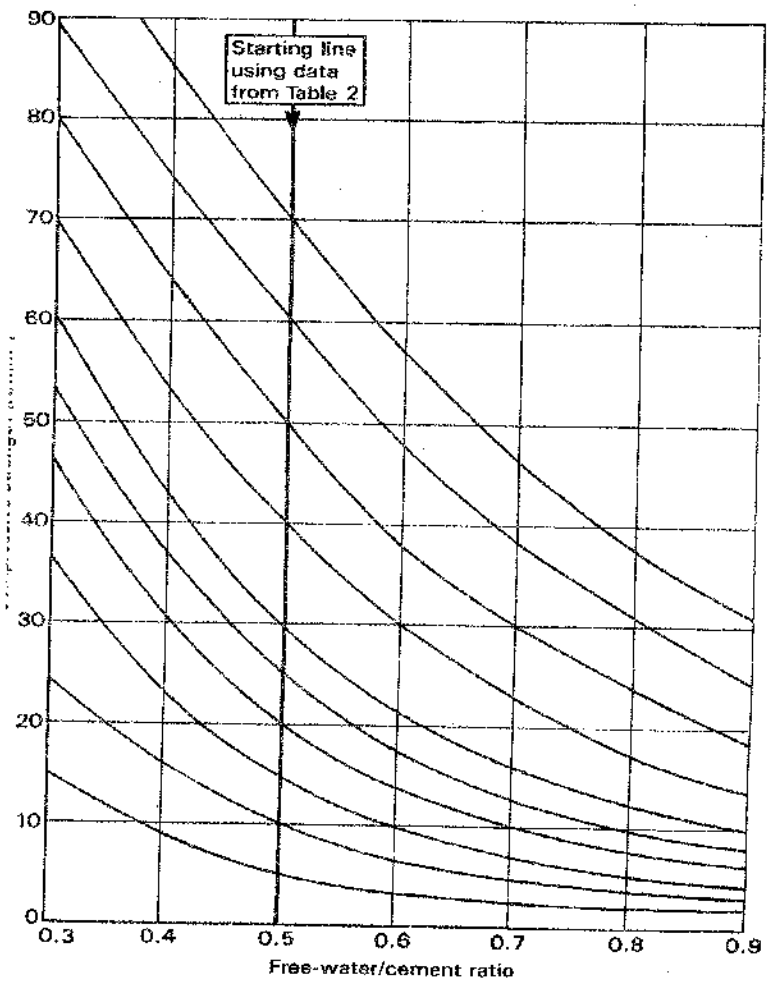

Figure 2
Relationship between compressive strength and free-water/cement ratio

Table 2 Approximate compressive strengths (N/mm²) of concrete mixes made with a free-water/cement ratio of 0.5

Cement strength class	Type of coarse aggregate	Compressive strengths (N/mm²) Age (days) 3	7	28	91
42.5	Uncrushed	22	30	42	49
	Crushed	27	36	49	56
52.5	Uncrushed	29	37	48	54
	Crushed	34	43	55	61

Table 3 Approximate free-water contents (kg/m³) required to give various levels of workability

Slump (mm)		0–10	10–30	30–60	60–180
Vebe time (s)		>12	6–12	3–6	0–3
Maximum size of aggregate (mm)	Type of aggregate				
10	Uncrushed	150	180	205	225
	Crushed	180	205	230	250
20	Uncrushed	135	160	180	195
	Crushed	170	190	210	225
40	Uncrushed	115	140	160	175
	Crushed	155	175	190	205

Note: When coarse and fine aggregates of different types are used, the free-water content is estimated by the expression:

$$\tfrac{2}{3} W_f + \tfrac{1}{3} W_c$$

where W_f = free-water content appropriate to type of fine aggregate
and W_c = free-water content appropriate to type of coarse aggregate.

Table 9 Approximate free-water contents required to give various levels of workability

Part A: Portland cement concrete

Slump (mm)		0–10	10–30	30–60	60–180
Vebe time (s)		>12	6–12	3–6	0–3
Maximum size of aggregate (mm)	Type of aggregate	Water content (kg/m³)			
10	Uncrushed	150	180	205	225
	Crushed	180	205	230	250
20	Uncrushed	135	160	180	195
	Crushed	170	190	210	225
40	Uncrushed	115	140	160	175
	Crushed	155	175	190	205

Part B: Portland cement/pfa concrete

Slump (mm)	0–10	10–30	30–60	60–180
Vebe time (s)	>12	6–12	3–6	0–3
Proportion, *p*, of pfa to cement plus pfa (%)	Reductions in water content (kg/m³)			
10	5	5	5	10
20	10	10	10	15
30	15	15	20	20
40	20	20	25	25
50	25	25	30	30

Table 10 Approximate compressive strengths of Portland cement/pfa made with a W/(C + 0.30F) ratio of 0.5

Cement strength class	Type of coarse aggregate	Compressive strength at 28 days (N/mm²)
42.5	Uncrushed	42
	Crushed	49
52.5	Uncrushed	48
	Crushed	55

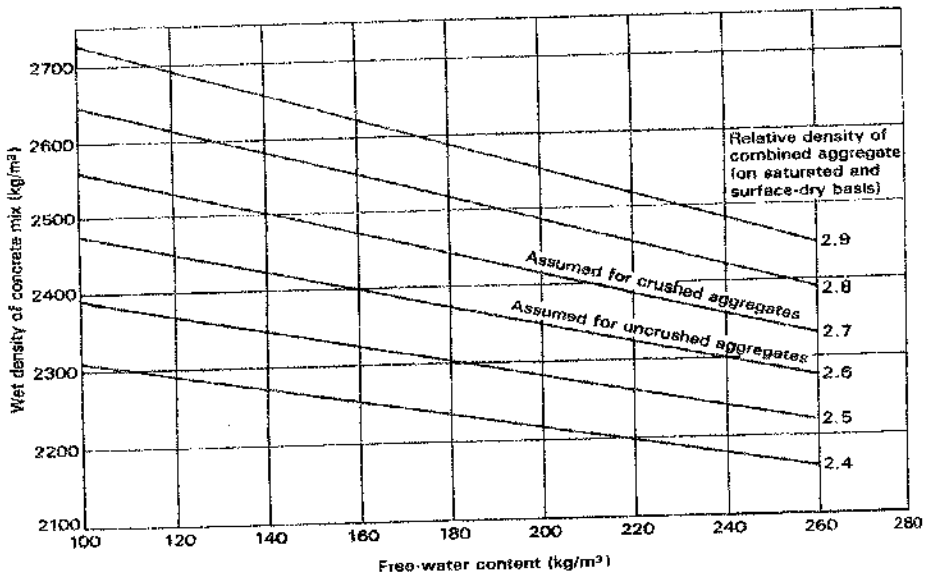

Figure 3 Estimated wet density of fully compacted concrete

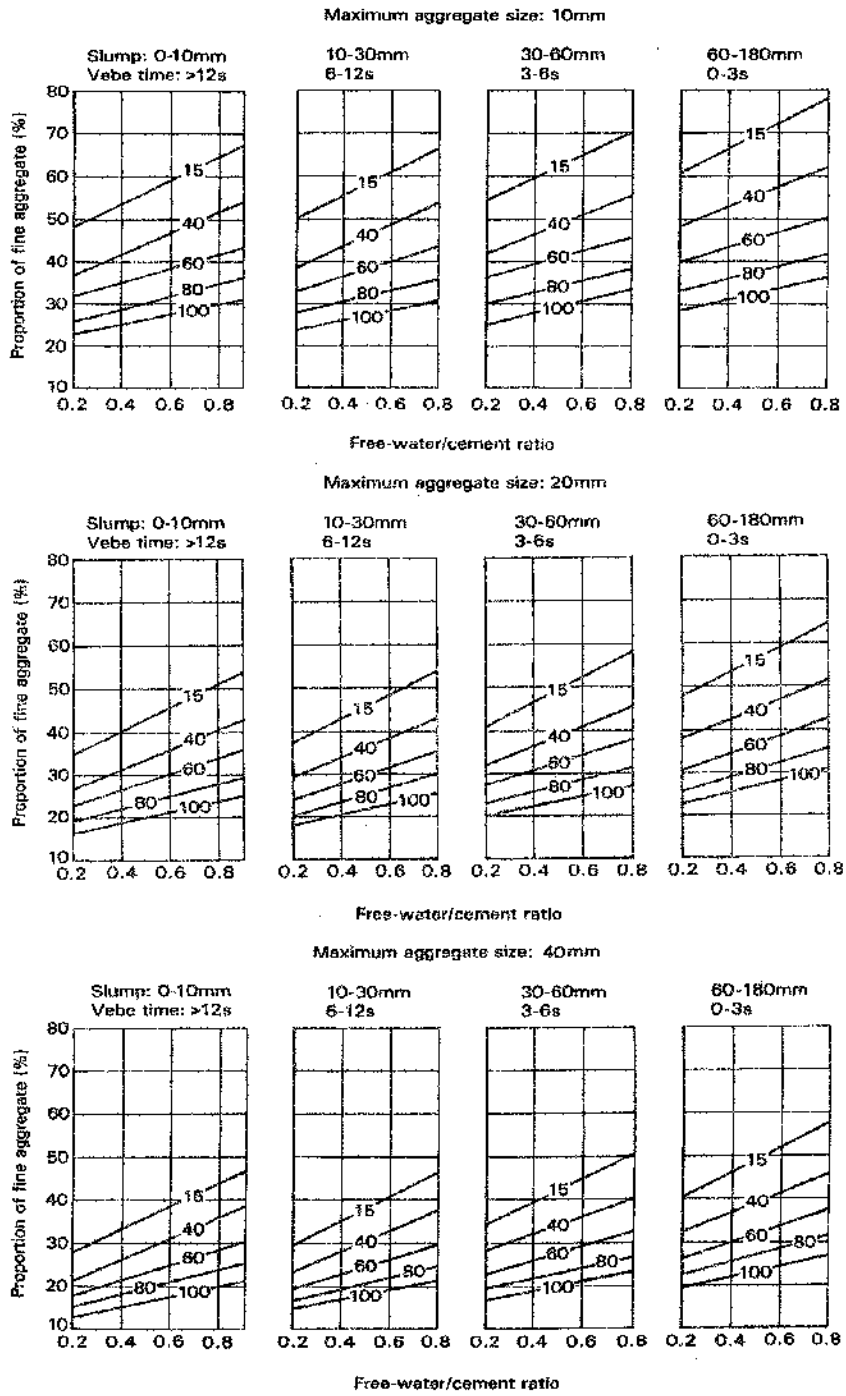

Figure 4 Recommended proportions of fine aggregate according to percentage passing 600 µm sieve